

Mukul Pal

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Experience

4 Way Technologies

Jul. 2025 – Feb. 2026

Software Development Intern

Delhi, India

- Co-led end-to-end development of an **OTT platform MVP** from 0 to 1 as part of a 2-person team, owning both backend and frontend independently
- Built and deployed **RESTful APIs** using **Next.js** to handle video streaming, user management, and subscription workflows
- Integrated **MUX** for adaptive video streaming and **AWS S3** for scalable media storage and delivery
- Developed the mobile client using **React Native**, enabling cross-platform video playback on iOS and Android
- Contributed to a **Conversational AI** product as a frontend developer; built responsive UI components using **React** and **TypeScript**

Projects

Devix | *Node.js, TypeScript, Socket.IO, Docker, PostgreSQL, AWS S3, React*

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- Built a **browser-based cloud IDE** where each project gets an **isolated Docker container** with a live Monaco editor, real file tree, and an integrated bash terminal all synced over **WebSockets**
- Engineered a **bidirectional terminal relay** using Socket.IO's **/shell namespace**, piping xterm.js keystrokes directly to a Docker exec stream and streaming container stdout back to the browser in real time
- Implemented a **hybrid storage layer** active project files are hot-cached on disk and volume-mounted into containers for low-latency I/O, then synced to **S3-compatible blob storage** on idle.
- Built an **IdleContainerReaper** background service that tracks per-project connection ref-counts, syncs files to S3, and tears down containers after a **2-minute grace period** preventing cold starts on accidental disconnects
- Designed a **reference-counted file watcher** shared across all users in a project room one watcher per project regardless of connection count, with **500ms debounce** on tree updates and a background **WatcherReaper** to reclaim leaked inotify descriptors

ResumeAI | *Node.js, React, PostgreSQL, BullMQ, Redis, OpenAI API*

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- Built an **AI-powered resume screening platform** that extracts structured data from PDFs and scores candidate-JD fit using **LLM analysis**, with output schema validation and a self-correcting retry loop for malformed responses
- Designed an **async job queue** using BullMQ and Redis, reducing perceived API latency from **8-12 seconds to under 100ms** by decoupling PDF extraction, LLM inference, and result persistence into background workers
- Implemented a **two-layer token optimisation pipeline** structured resume extraction on first upload and SHA-256 based JD caching cutting per-analysis LLM token consumption by **60%** on repeat submissions
- Engineered **PDF ingestion** with OCR fallback via Tesseract for scanned resumes, magic-byte file validation, and prompt injection sanitisation on JD text inputs

Skills

Languages: JavaScript, TypeScript

Frameworks & Libraries: Node.js, Express.js, React.js, Next.js, React Native

Databases: MongoDB, PostgreSQL, Prisma ORM, Mongoose

Cloud & DevOps: AWS S3, Docker, CI/CD, GitHub Actions

Tools: Git, GitHub, Postman, REST APIs, WebSockets

Education

Dr. APJ Abdul Kalam Technical University

Bachelor of Technology in Computer Science

Sept. 2021 – Sept. 2025

Ghaziabad, Uttar Pradesh